AEIC mechanism: from public contract to auditable collective decision

Observable actions connect individual follower choices, multi-round collaboration, and mechanism-level evaluation.

TASK / QUESTION — x + answer space

PUBLIC CONTRACT — reward · cost · policy

PANEL CONFIG — agents · rounds · seed

INFORMATION RULE — who observes what, when

A. FOLLOWER MICRO-ARCHITECTURE

AGENT 1 — analyst

offer m → action a_i → AEIC report r_i

OBSERVE | CHOOSE | EXECUTE | REPORT

AGENT 2 — skeptic

offer m → action a_i → AEIC report r_i

OBSERVE | CHOOSE | EXECUTE | REPORT

AGENT 3 — verifier

offer m → action a_i → AEIC report r_i

OBSERVE | CHOOSE | EXECUTE | REPORT

B. LEADER–FOLLOWER CONTROL PLANE

① Leader commitment
render visible credit–cost offer before generation

② Action router
declared action selects an enforceable resource path

LOW — short | AEIC — full | VERIFY — audit | IMITATE — peer

③ STRUCTURED REPORT INTERFACE

A Assumption | E Evidence | I Inference | C Conclusion

④ Settlement
quality credit − action cost

⑤ Persistent state
p_i · trace · tokens · latency

Round controller
synchronous barrier · rollback · advance

h_i^{t+1}

reports + policy

C. MULTI-AGENT COLLABORATION

INFORMATION ENVIRONMENTS

Independent | Evidence | Social | Copy | Counter-E

SYNCHRONOUS DELIBERATION

R1 → R2 → R3

t reads permitted reports from t−1; same-round isolation

Majority aggregation
explicit C fields → final decision

Behavior
effort · flips

Decision
accuracy · pairs

Cost / audit
tokens · failures

failure path: parse / API error → retry / rollback / fallback

Follower response — high-effort action

Decision quality — paired accuracy

Resource cost — tokens / calls

Validity boundary — held-out backend